



# Sébastien MARTINET

10/07/1996  
French

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## Languages

|                 |         |
|-----------------|---------|
| English         | ● ● ● ● |
| French (native) | ● ● ● ● |
| Spanish         | ● ● ● ○ |
| Italian         | ● ● ● ○ |

## Numerical tools

|            |         |
|------------|---------|
| Python     | ● ● ● ● |
| Fortran    | ● ● ● ● |
| C/C++      | ● ● ● ○ |
| R          | ● ● ● ○ |
| hTl5 & CSS | ● ● ● ○ |

## Machine Learning & Data Processing

|              |         |
|--------------|---------|
| Pandas       | ● ● ● ● |
| SQL          | ● ● ● ○ |
| scikit-learn | ● ● ● ○ |
| PyTorch      | ● ● ● ○ |
| TensorFlow   | ● ○ ○ ○ |
| PySpark      | ● ● ● ○ |

## Education

- 2022 PhD in Astrophysics**  
University of Geneva,  
Switzerland
- 2018 Master in Astrophysics**  
Université Grenoble-Alpes,  
France
- 2016 Bachelor Fundamental Physics**  
Université Pierre et Marie  
Curie, Paris VI, France

## Experience

- Oct. 2022 – Present. **PostDoctoral Researcher in Nuclear Astrophysics** Université Libre de Bruxelles, BELGIUM
- Sept. 2022 – Oct. 2022. **PostDoctoral Researcher in Stellar Astrophysics** University of Geneva, SWITZERLAND
- Sept. 2018 – Aug. 2022. **Doctoral Researcher in Stellar Astrophysics** University of Geneva, SWITZERLAND

## Data Science Projects

**Statistical Modeling & Scalable Data Pipelines** [Python](#), [Fortran](#), [HPC](#), [Git](#) (7 publications - 3 first-author)

- Built a scalable pipeline to run and analyze large-scale simulations (10k+ runs) using legacy Fortran code wrapped in Python; automated deployment on HPC clusters.
- Developed and applied a statistical framework to quantify and propagate uncertainties, enabling robust predictions from high-dimensional data.
- Delivered reproducible workflows and end-to-end tooling to extract actionable insights from complex simulation output.
- Collaborated with multidisciplinary teams to translate technical results into usable observables across domains.

**Enhancing Legacy Software with Modern Data Science** [Python](#), [Fortran](#), [Data Viz](#), [CLI Tools](#) (6 publications - 2 first-author )

- Led the design and integration of new physics modules into a legacy simulation code, improving performance and expanding capabilities for broader use cases.
- Modernized the data analysis interface with Python tooling (NumPy, pandas, seaborn), improving maintainability and enabling rapid visual exploration.
- Developed robust CLI tools and notebooks for data wrangling and visualization, enhancing reproducibility and team collaboration.
- Demonstrated autonomy and leadership in driving complex projects from concept to publication.

Other projects (55 collaborators, 19 institutes, 11 countries) include: integrating models with large observational datasets, analyzing experimental data to improve simulations, publishing data products, and building scientific data platforms.

## Selected Publications

Martinet, S. et al. (2025) Statistical framework for nuclear parameter uncertainties in nucleosynthesis modeling of r- and i-process.  
doi: [2025EPJA...61...48M](#)

Martinet, S. et al. (2024) The intermediate neutron capture process: IV. Impact of nuclear model and parameter uncertainties.  
doi: [2024A&6A...684A...8M](#)

[Full list of publications](#) / [Full list of conferences](#)  
(22 papers incl. 9 first author) / (15 talks in 10 different countries)